

WSI Viewer Release Cheat Sheet

Development -> Staging -> Production rehearsal -> Production -> Tag

Environments

Environment	Port	Runtime / identity	Data boundary
Development	8081	Live source; red banner	Deidentified development images and annotations
Staging	8082	Candidate JAR; yellow banner	Deidentified staging images and annotations
Rehearsal	8083	Exact staging JAR; production mode; loopback only	Deidentified production-marked images; rehearsal annotations
Production	8080	Frozen validated JAR; no banner	Authorized clinical images and production annotations

Normal release

One-command monitored workflow

```
./ops/wsi-release cycle --step
./ops/wsi-release cycle
./ops/wsi-release cycle --dry-run
./ops/wsi-release cycle --resume --step
```

Recommended: cycle --step displays every material action. Enter runs, p repeats, and q stops safely.

Automated phases

1. Repository/environment/production preflight
2. Development tests, restart and verification
3. Publish tested feature commit
4. Build/install/verify staging
5. Copy exact JAR to isolated rehearsal
6. Final identity, health and backup-space preflight
7. Verified backup then production promotion
8. Production QC and optional tag

Exact human gates

```
DEVELOPMENT-PASS
STAGING-PASS
REHEARSAL-PASS
PROMOTE
PRODUCTION-PASS
TAG (or SKIP)
```

Browser success is never inferred. Production is not stopped until its complete backup verifies. Failure prints exact rollback guidance; rollback is never automatic.

State and detailed log

```
.runtime/run/release-cycle.state
.runtime/log/cycle-<id>.log
```

Every browser gate: banner/title, login, image discovery/open/switch, tiles/pan/zoom, channel/display controls, full annotation naming/edit/delete/persistence and global Show/Hide, exports, isolated annotations, clean console/network, and no sustained delay.

Troubleshooting and Recovery

Status and verification

```
wsi production status
wsi staging status
wsi rehearsal status
wsi development status
./ops/wsi-release status
./ops/wsi-release verify staging
./ops/wsi-release verify rehearsal
./ops/wsi-release verify production
```

Logs

```
wsi production logs
wsi staging logs
wsi rehearsal logs
wsi development logs
```

Press **Control-C** to stop following a log.

Rollback

```
./ops/wsi-release history
./ops/wsi-release rollback --step
./ops/wsi-release verify production
```

Rollback restores the prior JAR, metadata, checksum and configuration. It **never restores or overwrites annotations**.

Troubleshooting modes

```
./ops/wsi-release cycle --resume --verbose
./ops/wsi-release stage --dry-run
./ops/wsi-release stage --step --verbose
./ops/wsi-release rehearse --dry-run
./ops/wsi-release promote --dry-run
./ops/wsi-release history
```

--dry-run: plan only. **--step**: Enter runs, p repeats, q exits safely. Use `./ops/wsi-release` if the installed command is missing. Never bypass PROMOTE or ROLLBACK.

Stop conditions

- Dirty Git tree or failed required test
- Git HEAD differs from staging commit
- Recorded/calculated SHA-256 differ
- Staging and rehearsal differ
- Environment identity, marker, root or port mismatch
- Production backup validation fails
- New 403, 500, unhandled JavaScript error, lost annotation, blank image or routine delay

Do not repair a mismatch by editing build metadata or checksum files.

Marker rule

Identity	Required marker
development	.wsi-environment-development
staging	.wsi-environment-staging
production / rehearsal	.wsi-environment-production

Timing: Image-switch total ends at OpenSeadragon open and may not include all visible tiles. Use browser Network timing and server logs for repeatable delays.

Privacy: Never copy production slides into development, staging or rehearsal. Verify filenames, metadata, embedded labels, thumbnails, macro images and associated files. Never synthesize a missing embedded label/thumbnail from diagnostic pixels.

Exactly one environment marker must exist. A missing, multiple or cross-environment marker prevents startup before the server accepts requests.